

HW 4

Given Grammar:

$A \rightarrow BAB \mid B \mid \varepsilon$

$B \rightarrow 00 \mid \varepsilon$

Step 1: Add New Start Symbol

$S_0 \rightarrow A$

$A \rightarrow BAB \mid B \mid \varepsilon$

$B \rightarrow 00 \mid \varepsilon$

Step 2: Remove ε -productions

Nullable variables: A, B

Updated productions after ε -removal:

$S_0 \rightarrow A \mid \varepsilon$

$A \rightarrow BAB \mid AB \mid BA \mid BB \mid A \mid B$

$B \rightarrow 00$

Step 3: Remove Unit Productions

Remove unit rules: $S_0 \rightarrow A$, $A \rightarrow B$, $A \rightarrow A$

Resulting grammar:

$S_0 \rightarrow BAB \mid AB \mid BA \mid BB \mid 00 \mid \varepsilon$

$A \rightarrow BAB \mid AB \mid BA \mid BB \mid 00$

$B \rightarrow 00$

Step 4: Convert to Chomsky Normal Form

Introduce $U \rightarrow 0$

$00 \rightarrow UU$

Introduce $E1 \rightarrow AB$

$BAB \rightarrow B E1$

Final CNF Grammar

$S0 \rightarrow B E1 \mid AB \mid BA \mid BB \mid UU \mid \varepsilon$

$A \rightarrow B E1 \mid AB \mid BA \mid BB \mid UU$

$B \rightarrow UU$

$E1 \rightarrow AB$

$U \rightarrow 0$